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MODULE IV

COMPLEX VARIABLE - INTEGRATION

(Text 2 - Relevant topics from 14.1, 14.2, 14.3, 14.4, 15.4)

Syllabus : Complex integration, Line integrals in the complex plane, Basic properties, First evaluation method - indefinite integration and substitution of limit, second evaluation method - use of a representation of a path, Contour integrals, Cauchy integral theorems (without proof) on simply connected domain, Cauchy integral theorems (without proof) on multiply connected domain, Cauchy integral formula (without proof) for derivatives of an analytic function, Taylor's series and Maclaurin's series.

Line integrals in the complex plane

Complex definite integrals are called line integrals and it is defined in terms of the values of a complex valued function $f(z)$ along a given curve (arc) C extending from a point $z = z_1$ to a point $z = z_2$ in the complex plane, written as

$$\int_C f(z) dz$$

The curve C is called the path of integration. C may be represented parametrically by

$$z(t) = x(t) + i y(t) \quad (a \leq t \leq b)$$

[The sense of increasing t is called the positive sense on C and we say that C is oriented by $z(t)$]

The curve C is assumed to be a smooth curve
ie, C has a continuous and non-zero derivative

$$z'(t) = \frac{dz}{dt} = x'(t) + iy'(t) \text{ at each point}$$

If C is a closed curve, then the line integral is called contour integral denoted as

$$\oint_C f(z) dz.$$

General assumption: All paths of integration for complex line integrals are assumed to be piecewise smooth, that is, they consist of finitely many smooth curves joined end to end.

Basic Properties

1. Linearity: Integration is a linear operation.
ie, we can integrate sums term by term and can take out constant factors from under the integral sign.

$$\int_C [k_1 f_1(z) + k_2 f_2(z)] dz = k_1 \int_C f_1(z) dz + k_2 \int_C f_2(z) dz$$

2. Sense reversal: $\int_{z_0}^{z_1} f(z) dz = - \int_{z_1}^{z_0} f(z) dz$

3. Partitioning of path: $\int_C f(z) dz = \int_{C_1} f(z) dz + \int_{C_2} f(z) dz$

where curve C consists of the curves C_1 and C_2 such that the initial point of C_2 being the terminal point of C_1

Existence of the complex line integral

$f(z)$ is continuous and C is piecewise smooth imply the existence of the line integral $\int_C f(z) dz$ or $\oint f(z) dz$.

Evaluation of Line integrals

I) First Evaluation method: (Indefinite integration and substitution of limits)

Theorem 1: (Indefinite integration of analytic functions)

Let $f(z)$ be analytic in a simply connected* domain D . Then there exists an indefinite integral (antiderivative) of $f(z)$ in D , that is, an analytic function $F(z)$ such that

$$F'(z) = f(z) \text{ in } D.$$

and for all paths in D joining two points z_0 and z_1 in D we have

$$\int_{z_0}^{z_1} f(z) dz = F(z_1) - F(z_0) \quad \left[\text{where } F'(z) = f(z) \right]$$

(Note that we can write z_0 and z_1 instead of C , since we get the same value for all those C from z_0 to z_1)

* Note: Simply connected domain

A domain D is called simply connected if every simple closed curve (closed curve without self-intersections) encloses only points of D .

eg: interior of a circle, rectangle, triangle, ellipse

Note: First evaluation method is suitable for analytic functions only.

Problems : Evaluate

i) $\int_0^{1+i} z^2 dz$

Ans: $f(z) = z^2$

$$F(z) = \int f(z) dz = \int z^2 dz = \frac{z^3}{3} + C$$

$F(z)$ is an antiderivative of $f(z)$ i.e., $F'(z) = f(z)$ and $f(z) = z^2$ is analytic in the complex plane

$\therefore$ by theorem 1 (by first evaluation method),

$$\int_0^{1+i} z^2 dz = \left[\frac{z^3}{3} \right]_0^{1+i} = \frac{1}{3} [(1+i)^3 - 0]$$

$$= \frac{1}{3} [1 + 3i - 3 - i - 0]$$

$$= \frac{1}{3} (2i - 2) = \frac{2}{3} (i - 1)$$

$$= \underline{\underline{-\frac{2}{3} + \frac{2}{3}i}}$$

ii) $\int_{-\pi i}^{\pi i} \cos z dz$

Ans: $f(z) = \cos z$

$$F(z) = \int f(z) dz = \int \cos z dz = \sin z + C$$

$F(z)$ is an antiderivative of $f(z)$ i.e., $F'(z) = f(z)$ and $f(z) = \cos z$ is analytic in the complex plane

$\therefore$ by first evaluation method

$$\int_{-\pi i}^{\pi i} \cos z dz = [\sin z]_{-\pi i}^{\pi i}$$

$$= \sin \pi i - \sin(-\pi i)$$

$$= \sin \pi i + \sin \pi i = 2 \sin \pi i$$

$$= \underline{\underline{2i^0 \sinh \pi}}$$

$$[\text{Result : } \sin(iy) = i \sinh y]$$

$$\text{iii)} \int_{8+\pi i}^{8-3\pi i} e^{z/2} dz$$

Ans: $f(z) = e^{z/2}$

$$F(z) = \int f(z) dz = \int e^{z/2} dz = \frac{e^{z/2}}{1/2} = 2e^{z/2} + C$$

and $f(z) = e^{z/2}$ is analytic in the complex plane

$\therefore$ by first evaluation method

$$\int_{8+\pi i}^{8-3\pi i} e^{z/2} dz = \left(\frac{e^{z/2}}{1/2} \right)_{8+\pi i}^{8-3\pi i} = 2 \left(e^{z/2} \right)_{8+\pi i}^{8-3\pi i}$$

$$= 2 \left[e^{\frac{8-3\pi i}{2}} - e^{\frac{8+\pi i}{2}} \right]$$

$$= 2 \left[e^4 \cdot e^{-\frac{3\pi i}{2}} - e^4 \cdot e^{\frac{\pi i}{2}} \right]$$

$$= 2e^4 \left[\cos\left(\frac{3\pi}{2}\right) - i \sin\left(\frac{3\pi}{2}\right) - \left(\cos\left(\frac{\pi}{2}\right) + i \sin\left(\frac{\pi}{2}\right) \right) \right]$$

$$= 2e^4 \left[0 - i(-1) - 0 - i(1) \right]$$

$$= 2e^4 [i - i] = \underline{\underline{0}}$$

$\therefore$ [Results: $e^{i\theta} = \cos \theta + i \sin \theta$; $e^{-i\theta} = \cos \theta - i \sin \theta$]

$$\text{iv)} \int_{-i}^i \frac{dz}{z}$$

Ans: $f(z) = \frac{1}{z}$ analytic except at $z=0$

$$F(z) = \int f(z) dz = \int \frac{1}{z} dz = \ln z + C$$

and $F(z)$ is analytic except at 0 and on the negative real axis.

$\therefore$ by first evaluation method

$$\int_{-i}^i \frac{dz}{z} = (\ln z)_{-i}^i = \ln(i) - \ln(-i)$$

$$= \ln(e^{i\pi/2}) - \ln(e^{-i\pi/2})$$

$$= i\pi/2 - (-i\pi/2)$$

$$= i\pi/2 + i\pi/2 = 2 \frac{i\pi}{2}$$

$$= \underline{i\pi} \quad \left(\text{Here the domain } D \text{ is the complex plane without } 0 \text{ and the negative real axis} \right) \ln e^x = x$$

$$\text{Results: } i^0 = \cos \frac{\pi}{2} + i \sin \frac{\pi}{2}$$

$$= e^{i\pi/2}$$

$$-i^0 = \cos \pi - i \sin \pi$$

$$= e^{-i\pi/2}$$

II) Second Evaluation method

This method is not restricted to analytic functions but applies to any continuous complex function.

Theorem : (Integration by the use of the path)

Let C be a piecewise smooth path represented by $z = z(t) = x(t) + iy(t)$; $t \in [a, b]$

Let $f(z)$ be a continuous function on C . Then

$$\int_C f(z) dz = \int_a^b f(z(t)) z'(t) dt$$

Problems :

i) Evaluate $\int_0^{1+i} (x^2 - iy) dz$ along the paths

a) $y = x$

b) $y = x^2$

Ans: Along the line $y = x$

$$dy = dx \text{ so that } dz = dx + i dy = dx + i dx = (1+i) dx$$

$$\therefore \int_0^{1+i} (x^2 - iy) dz = \int_0^{1+i} (x^2 - iy) (dx + i dy)$$

$$= \int_0^1 (x^2 - ix) (1+i) dx$$

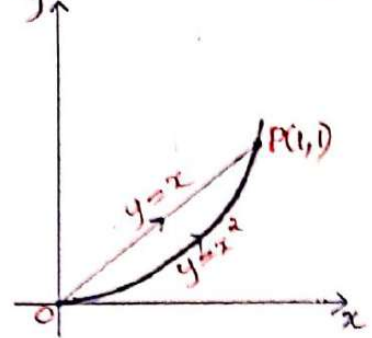
$$\begin{aligned} z &: (0,0) \rightarrow (1,1) \\ \Rightarrow x &: 0 \rightarrow 1 \\ y &: 0 \rightarrow 1 \end{aligned}$$

$$= (1+i) \int_0^1 (x^2 - ix) dx$$

$$= (1+i) \left[\frac{x^3}{3} - i \frac{x^2}{2} \right]_0^1$$

$$= (1+i) \left(\frac{1}{3} - \frac{i}{2} \right)$$

$$= \frac{(1+i)(2-3i)}{6} = \underline{\underline{\frac{5}{6} - \frac{1}{6}i}}$$



b) Along the parabola $y=x^2$, $dy=2x dx$ so that

$$\therefore dz = dx + i dy = dx + 2ix dx = (1+2ix) dx$$

x varies from 0 to 1 [$\because z : (0,0) \rightarrow (1,1)$
 $\rightarrow x : 0 \rightarrow 1$]

$$\therefore \int_0^{1+i} (x^2 - iy) dz = \int_0^1 (x^2 - ix^3) (1+2ix) dx$$

$$= (1-i) \int_0^1 x^2 (1+2ix) dx$$

$$= (1-i) \int_0^1 (x^2 + 2ix^3) dx$$

$$= (1-i) \left(\frac{x^3}{3} + 2i \frac{x^4}{4} \right)_0^1$$

$$= (1-i) \left(\frac{1}{3} + \frac{i}{2} \right) = \frac{(1-i)(2+3i)}{6}$$

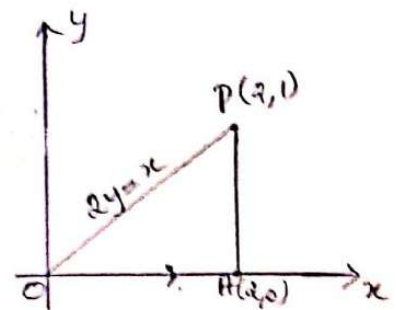
$$= \underline{\underline{\frac{5}{6} + \frac{1}{6}i}}$$

Note : The values of the integral are different along the two paths.

2) Evaluate $\int_0^{2+i} (\bar{z})^2 dz$, along

a) the real axis to 2 and then vertically to $2+i$

b) along the line $2y=x$



Ans: $(\bar{z})^2 = (x-iy)^2 = x^2 - y^2 - i2xy$

a) Along the path OAP where A is (2,0) and P is (2,1)

$$\therefore \int_0^{2+i} (\bar{z})^2 dz = \int_{OA} (x^2 - y^2 - i2xy) dz + \int_{AP} (x^2 - y^2 - i2xy) dz \quad (1)$$

Now, along OA

$y=0 \Rightarrow dz = dx + i dy = dx$ and x varies from 0 to 2

Along AP

$x=2 \Rightarrow dz = dx + i dy = 0 + i dy = i dy$ and y varies from 0 to 1

$$\begin{aligned} \therefore (1) \Rightarrow \int_0^{2+i} (\bar{z})^2 dz &= \int_0^2 x^2 dx + \int_0^1 (4 - y^2 - i4y) i dy \\ &= \left(\frac{x^3}{3} \right)_0^2 + \left(4iy - \frac{iy^3}{3} + 4\frac{y^2}{2} \right)_0^1 \\ &= \frac{8}{3} + 4i - \frac{i}{3} + 2 \\ &= \underline{\underline{\frac{14}{3} + \frac{11}{3}i}} \end{aligned}$$

b) Along the line $2y = x$, $2dy = dx$

$$\therefore dz = dx + i dy = 2dy + i dy = (2+i) dy$$

and y varies from 0 to 1

$$\begin{aligned} \therefore \int_0^{2+i} (\bar{z})^2 dz &= \int_0^1 (x^2 - y^2 - i2xy) (2+i) dy \\ &= \int_0^1 (4y^2 - y^2 - i2 \cdot 2y \cdot y) (2+i) dy \\ &= \int_0^1 (3y^2 - i4y^2) (2+i) dy \\ &= \int_0^1 (3 - 4i) y^2 (2+i) dy \end{aligned}$$

$$\begin{aligned}
 &= (3-4i)(2+i) \int_0^1 y^2 dy \\
 &= (10-5i) \left(\frac{y^3}{3} \right)_0^1 = \underline{\underline{\frac{10}{3} - \frac{5}{3}i}}
 \end{aligned}$$

3. Evaluate $\int_C \operatorname{Re}(z) dz$ where C is the shortest path from $1+i$ to $5+5i$.

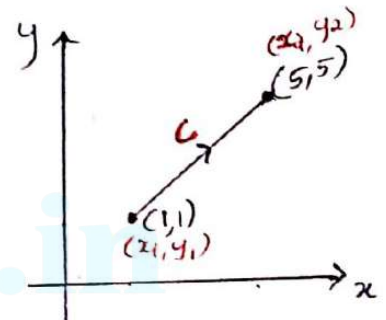
Ans: The shortest path from $1+i$ to $5+5i$ is the straight line joining the points $(1,1)$ (Corresponds to $1+i$) and $(5,5)$ (Corresponds to $5+5i$).

$\therefore$ Eqn of the path of integration C is given by

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1}$$

$$\Rightarrow \frac{x-1}{4} = \frac{y-1}{4}$$

$$\Rightarrow x = y$$



$$z = x+iy \Rightarrow \operatorname{Re}(z) = x$$

$$dz = dx + i dy = dx + i dx \quad [\because x=y \Rightarrow dx=dy]$$

and x varies from 1 to 5

$$\begin{aligned}
 \therefore \int_C \operatorname{Re}(z) dz &= \int_{1+i}^{5+5i} x dz = \int_{1+i}^{5+5i} x(dx + i dy) \\
 &= \int_1^5 x(dx + i dx) = \int_1^5 x(1+i) dx \\
 &= (1+i) \int_1^5 x dx \\
 &= (1+i) \left(\frac{x^2}{2} \right)_1^5 \\
 &= (1+i) \left(\frac{25}{2} - \frac{1}{2} \right) \\
 &= \underline{\underline{12(1+i)}}
 \end{aligned}$$

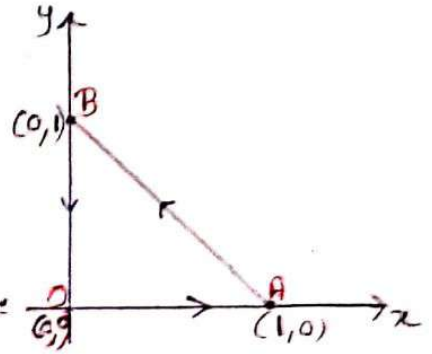
4. Evaluate $\oint \operatorname{Im}(z^2) dz$ counterclockwise around the triangle with vertices $0, 1, i$

Ans : $z = x + iy$

$$\therefore z^2 = (x + iy)^2 = x^2 - y^2 + i2xy$$

$$\Rightarrow \operatorname{Im}(z^2) = 2xy$$

Take $C : \triangle OAB$ in counterclockwise direction



Along OA : $y = 0 \Rightarrow dy = 0$
 $\therefore dz = dx + i dy = dx$
 x varies from 0 to 1

$$\therefore \int_{OA} \operatorname{Im}(z^2) dz = \int_{OA} 2xy dz = \int_0^1 0 dx = 0$$

Along AB : path of integration is given by straight line joining $(1,0)$ and $(0,1)$

$$\frac{x - x_1}{x_2 - x_1} = \frac{y - y_1}{y_2 - y_1}$$

$$\Rightarrow \frac{x - 1}{0 - 1} = \frac{y - 0}{1 - 0}$$

$$\Rightarrow 1 - x = y$$

$$\therefore dz = dx + i dy = dx - i dx$$

$$\begin{cases} y = 1 - x \\ dy = -dx \end{cases}$$

and x varies from 1 to 0

$$\begin{aligned} \therefore \int_{AB} \operatorname{Im}(z^2) dz &= \int_1^0 2x(1-x)(dx - i dx) \\ &= \int_1^0 (2x - 2x^2)(1-i) dx \\ &= (1-i) \left(2 \frac{x^2}{2} - 2 \frac{x^3}{3} \right) \Big|_1^0 \\ &= (1-i) \left(-1 + \frac{2}{3} \right) = -\frac{1}{3}(1-i) \end{aligned}$$

Along BO : $x=0 \Rightarrow dx=0$
 $dz = dx + i dy = i dy$

$$\therefore \int_{BO} \text{Im}(z^2) dz = \int_{BO} 2xy dz = \int_1^0 (0) i dy = 0$$

$$\begin{aligned} \therefore \oint_C \text{Im}(z^2) dz &= \int_{OA} \text{Im}(z^2) dz + \int_{AB} \text{Im}(z^2) dz + \int_{BO} \text{Im}(z^2) dz \\ &= 0 - \frac{1}{3}(1-i) + 0 \\ &= \underline{\underline{-\frac{1}{3}(1-i)}} \end{aligned}$$

5. Integrate $f(z) = x^2 + ixy$ from $A(1,1)$ to $B(2,4)$ along the curve $x=t, y=t^2$

Ans : Equation of the path of integration is given by

$$\left. \begin{aligned} x &= t, \quad y = t^2 \\ \therefore dx &= dt, \quad dy = 2t dt \end{aligned} \right\} \Rightarrow dz = dx + i dy = dt + i 2t dt$$

$$f(z) = f(z(t)) = x^2 + ixy = t^2 + i t^3$$

$$\begin{aligned} \therefore \int_A^B f(z) dz &= \int_{1+i}^{2+4i} (x^2 + ixy) dz && \begin{aligned} z &: (1,1) \rightarrow (2,4) \\ \Rightarrow x &: 1 \rightarrow 2 \\ \Rightarrow t &: 1 \rightarrow 2 \end{aligned} \\ &= \int_1^2 (t^2 + i t^3) (dt + i 2t dt) && [\because t=x] \\ &= \int_1^2 (t^2 + i t^3) (1 + i 2t) dt \\ &= \int_1^2 (t^2 + 3i t^3 - 2t^4) dt \\ &= \left(\frac{t^3}{3} + 3i \frac{t^4}{4} - 2 \frac{t^5}{5} \right)_1^2 \\ &= \frac{8}{3} + \frac{3i}{4} (16) - \frac{2}{5} (32) - \frac{1}{3} - \frac{3i}{4} + \frac{2}{5} \\ &= \underline{\underline{-\frac{151}{15} + \frac{45}{4} i}} \end{aligned}$$

6. Evaluate $\oint_C \frac{1}{z} dz$ where C is the circle $|z|=1$

Ans: The parametric equation of the circle $|z|=1$ is

$$x = x(t) = \cos t \quad ; \quad y = y(t) = \sin t$$

$$\therefore z = z(t) = x(t) + iy(t) = \cos t + i \sin t = e^{it}$$

$$\therefore C \text{ is given by } C: z = e^{it} \quad ; \quad 0 \leq t \leq 2\pi$$

$$\therefore \oint_C \frac{1}{z} dz = \int_0^{2\pi} \frac{1}{e^{it}} i e^{it} dt$$

$$= \int_0^{2\pi} i dt$$

$$= i(t)_0^{2\pi}$$

$$= \underline{\underline{2\pi i}}$$

$$\begin{array}{l} z = e^{it} \\ \frac{dz}{dt} = i e^{it} \\ \Rightarrow dz = i e^{it} dt \end{array}$$

7. Prove that

$$i) \oint_C \frac{dz}{z-a} = 2\pi i$$

$$ii) \oint_C (z-a)^n dz = 0 \quad [n \text{ is an integer } \neq -1]$$

where C is the circle $|z-a|=r$

Ans: The parametric equation of the circle is
 $z-a = re^{i\theta} \Rightarrow z = a + re^{i\theta} \quad (0 \leq \theta \leq 2\pi)$

$$\therefore dz = i r e^{i\theta} d\theta$$

$$i) \oint_C \frac{dz}{z-a} = \int_0^{2\pi} \frac{i r e^{i\theta} d\theta}{r e^{i\theta}} = i \int_0^{2\pi} d\theta = 2\pi i$$

$$ii) \oint_C (z-a)^n dz = \int_0^{2\pi} (r e^{i\theta})^n i r e^{i\theta} d\theta$$

$$= \int_0^{2\pi} r^n e^{in\theta} \cdot i r e^{i\theta} d\theta$$

$$= i r^{n+1} \int_0^{2\pi} e^{in\theta} \cdot e^{i\theta} d\theta$$

$$= i r^{n+1} \int_0^{2\pi} e^{i(n+1)\theta} d\theta$$

$$= i r^{n+1} \left[\frac{e^{i(n+1)\theta}}{i(n+1)} \right]_0^{2\pi}$$

$$= \frac{r^{n+1}}{n+1} \left[e^{i 2(n+1)\pi} - e^0 \right]$$

$$= \frac{r^{n+1}}{n+1} [1-1] \quad \left[\because e^{i 2(n+1)\pi} = \cos 2(n+1)\pi + i \sin 2(n+1)\pi \right]$$

$$= \underline{\underline{0}}$$

$$= 1 + i0$$

$$= 1$$

Home work

1. Evaluate $\int_C e^{\pi z} dz$ where C is any path from $z=0$ to $z=1+i$

2. Evaluate $\int_{-2}^{-2+i} (2+z)^2 dz$

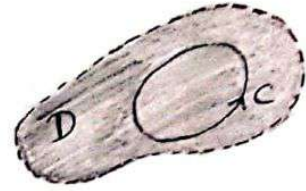
3. ST $\oint_C |z|^2 dz = -1+i$ where C is the rectangle with vertices $(0,0)$, $(1,0)$, $(1,1)$ & $(0,1)$ in anticlockwise direction

4. Evaluate $\int_C \bar{z} dz$ where C is the upper half of the circle $|z|=1$ from $z=-1$ to $z=1$.

Cauchy's Integral Theorem (Cauchy - Goursat Theorem)

If $f(z)$ is analytic in a simply connected domain D , then for every simple closed path C in D ,

$$\oint_C f(z) dz = 0$$



Note: The condition that the domain D be simply connected is essential.

Problems: i) Evaluate the following for any closed path C

i) $\oint_C e^z dz$ ii) $\oint_C \cos z dz$ iii) $\oint_C z^n dz$ ($n=0,1,2,\dots$)

Ans: i) $f(z) = e^z$ is analytic for all z in the complex plane

$\therefore \oint_C e^z dz = 0$ (by Cauchy's integral theorem)

ii) $f(z) = \cos z$ is analytic for all z in the complex plane

$\therefore \oint_C \cos z dz = 0$ (by Cauchy's integral theorem)

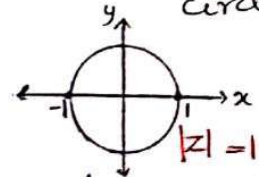
iii) $f(z) = z^n$ ($n=0,1,2,\dots$) are analytic in the entire complex plane.

$\therefore \oint_C z^n dz = 0$ (by Cauchy's integral theorem)

2) Evaluate i) $\oint_C \sec z dz$ ii) $\oint_C \frac{dz}{z^2+4}$ where C is the unit circle.

Ans: Here $C: |z|=1$

$f(z) = \sec z = \frac{1}{\cos z}$ is not analytic at $\cos z = 0$



$$\cos z = 0 \Rightarrow z = (2n+1)\frac{\pi}{2}; n=0, \pm 1, \pm 2, \dots$$

$$\Rightarrow z = \pm \frac{\pi}{2}, \pm \frac{3\pi}{2}, \dots$$

But all these points lie outside C ,
none lies on C or inside C

$$\therefore \oint_{|z|=1} z^n dz = 0 \quad (\text{by Cauchy's integral theorem}) \quad \left[\begin{array}{l} z = \pm \frac{\pi}{2} = \pm 1.57 \\ z = \pm \frac{3\pi}{2} = \pm 4.71 \\ z = \pm \frac{5\pi}{2} = \pm 7.85 \\ \dots \\ \rightarrow |z| > 1 \end{array} \right]$$

ii) $f(z) = \frac{1}{z^2+4}$ is not analytic at $z^2+4=0$
 $\Rightarrow z^2 = -4$
 $\Rightarrow z = \pm 2i$

But these points lie outside $C: |z|=1$

$$\therefore \oint_C \frac{1}{z^2+4} dz = 0 \quad (\text{by Cauchy's integral theorem}) \quad \left[\begin{array}{l} z = 2i = (0, 2) \\ z = -2i = (0, -2) \\ \text{both lie outside} \\ |z| = 1 \end{array} \right]$$

Note : Converse of Cauchy's theorem need not be true

Consider $\oint_C \frac{1}{z^2} dz$ where $C: |z|=1$

$$\begin{aligned} \therefore \oint_C \frac{1}{z^2} dz &= \int_0^{2\pi} \frac{1}{e^{i2\theta}} i e^{i\theta} d\theta \\ &= i \int_0^{2\pi} e^{-i\theta} d\theta \\ &= i \left(\frac{e^{-i\theta}}{-i} \right)_0^{2\pi} \\ &= - \left[e^{-i\theta} \right]_0^{2\pi} = - \left[e^{-i2\pi} - e^0 \right] \\ &= - [\cos 2\pi - i \sin 2\pi - 1] = - [1 - 0 - 1] = 0 \end{aligned} \quad \left[\begin{array}{l} z = e^{i\theta} \\ z^2 = e^{i2\theta} \\ \frac{dz}{d\theta} = e^{i\theta} i \\ \therefore dz = i e^{i\theta} d\theta \end{array} \right]$$

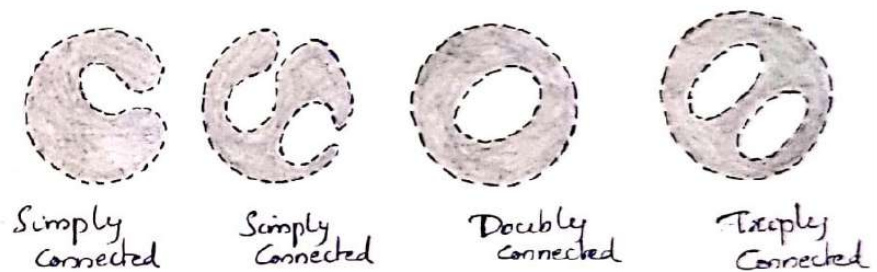
$$\Rightarrow \oint_C \frac{1}{z^2} dz = 0$$

But $f(z) = \frac{1}{z^2}$ is not analytic at $z=0$
 which is an interior point of $|z|=1$

$$\therefore \oint_C f(z) dz = 0 \not\Rightarrow f(z) \text{ is analytic.}$$

Definitions:

1. A **simple closed path** is a closed path that does not intersect or touch itself.
 eg: circle, a curve shaped like an 8
2. A **simply connected domain** D in the complex plane is a domain such that every simple closed path in D encloses only points of D .
 eg: The interior of a circle, ellipse, or any simple closed curve.
3. A domain that is not simply connected is called **multiply connected**.
 eg: An annulus, a disk without centre ($0 < |z| < 1$)



Cauchy's Integral Theorem for Multiply Connected Domains

Let C be a closed path and let $C_1, C_2, \dots, C_n$ be closed paths enclosed by C . Assume that no two of $C, C_1, C_2, \dots, C_n$ intersect, and no point interior to any C_i is interior to any other C_k . Let $f(z)$ be analytic on an open set containing C , each C_i and all points that are interior to C and exterior to each C_i . Then

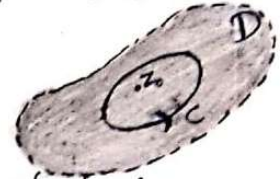
$$\oint_C f(z) dz = \oint_{C_1} f(z) dz + \oint_{C_2} f(z) dz + \dots + \oint_{C_n} f(z) dz$$

Orientation counter-clockwise on C and clockwise on each C_i

Cauchy's Integral Formula:

Let $f(z)$ be analytic in a simply connected domain D . Then for any point z_0 in D and any simple closed path C in D that encloses z_0

$$\oint_C \frac{f(z)}{z-z_0} dz = 2\pi i f(z_0)$$



the integration being taken counterclockwise.

$$\text{Alternatively, } f(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{z-z_0} dz$$

Problems

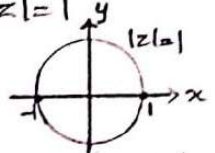
1. Evaluate $\oint_C \frac{e^z}{z} dz$ where C is $|z|=1$

Ans: Let $F(z) = \frac{e^z}{z}$ not analytic at $z=0$

The point $z=0$ lies inside the curve $C: |z|=1$

$$\therefore \oint_C \frac{e^z}{z} dz = \oint_C \frac{f(z)}{z-0} dz$$

where $f(z) = e^z$ is analytic inside C
 $z_0 = 0$ point inside C



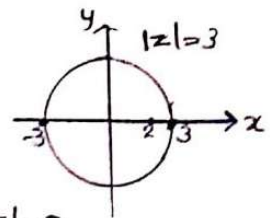
$$\begin{aligned} &= 2\pi i f(0), \text{ by Cauchy's integral formula} \\ &= 2\pi i e^0 \\ &= \underline{\underline{2\pi i}} \end{aligned}$$

2. Evaluate $\oint_C \frac{\cos \pi z}{z-2} dz$ where C is $|z|=3$

Ans: Let $F(z) = \frac{\cos \pi z}{z-2}$

not analytic at $z-2=0 \Rightarrow z=2$

The point $z=2$ lies inside the curve $|z|=3$



$$\therefore \oint_C \frac{\cos \pi z}{z-2} dz = \oint_C \frac{f(z)}{z-2} dz$$

where $f(z) = \cos \pi z$ is analytic inside C .

$$\begin{aligned}
 &= 2\pi i f(2) \text{ , by Cauchy's integral formula.} \\
 &= 2\pi i \cos 2\pi = 2\pi i (1) \\
 &= \underline{2\pi i}
 \end{aligned}$$

3. Evaluate $\oint_C \frac{z^3-6}{2z-i}$ where C is any contour enclosing $z_0 = i/2$

Ans : Let $F(z) = \frac{z^3-6}{2z-i}$ not analytic at $2z-i=0$
 $\Rightarrow z = i/2$

$\therefore$ by Cauchy's integral formula

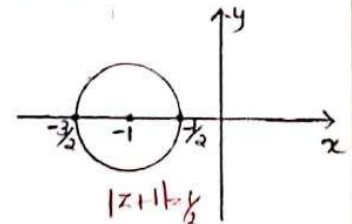
$$\begin{aligned}
 \oint_C \frac{z^3-6}{2z-i} dz &= \oint_C \frac{f(z)}{z-i/2} \text{ where } f(z) = \frac{z^3-6}{2} \\
 &= 2\pi i f(i/2) \quad \text{analytic inside } C \text{ (polynomial fun)} \\
 &= 2\pi i \left[\frac{z^3-6}{2} \right]_{z=i/2} \\
 &= 2\pi i \left[\frac{(i/2)^3-6}{2} \right] \\
 &= 2\pi i \left[\frac{i^3/8-6}{2} \right] \\
 &= 2\pi i \left[\frac{-i/8-6}{2} \right] \\
 &= \pi i (-i/8-6) = \underline{\underline{\frac{\pi}{8} - 6\pi i}}
 \end{aligned}$$

4. Evaluate $\int_C \frac{e^z}{z+1} dz$ where C is $|z+1| = 1/2$

Ans : Let $F(z) = \frac{e^z}{z+1}$ not analytic at $z+1=0 \Rightarrow z=-1$

The point $z=-1$ lies inside $|z+1|=1/2$

$$[\because z=-1 \Rightarrow |z+1| = |-1+1| = 0 < 1/2]$$



$$\therefore \oint_C \frac{e^z}{z+1} dz = \oint_C \frac{f(z)}{z+1} dz \text{ where } f(z) = e^z \text{ is analytic inside } C \\
 z_0 = -1 \text{ inside } C$$

$$= 2\pi i \int (-1), \text{ by Cauchy's integral formula.}$$

$$= \underline{2\pi i e^{-1}}$$

5. Evaluate $\oint_C \frac{z^2 + 2z + 3}{z^2 - 1} dz$ where C is $|z - 1| = 1$

Ans: Let $F(z) = \frac{z^2 + 2z + 3}{z^2 - 1}$ not analytic at $z^2 - 1 = 0 \Rightarrow z = \pm 1$

The point $z = 1$ lies inside $C: |z - 1| = 1$ and $z = -1$ lies outside C .

$$\left[\begin{array}{l} \because z = 1 \Rightarrow |z - 1| = |1 - 1| = 0 < 1 \Rightarrow z = 1 \text{ inside } C \\ z = -1 \Rightarrow |-1 - 1| = |-2| = 2 > 1 \Rightarrow z = -1 \text{ outside } C \end{array} \right]$$

$$\therefore \oint_C \frac{z^2 + 2z + 3}{z^2 - 1} dz = \oint_C \frac{z^2 + 2z + 3}{(z + 1)(z - 1)} dz$$

$$= \oint_C \frac{f(z)}{z - 1} dz$$

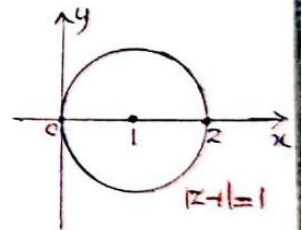
where $f(z) = \frac{z^2 + 2z + 3}{z + 1}$ is analytic inside C

$$= 2\pi i f(1)$$

$$= 2\pi i \times \left[\frac{z^2 + 2z + 3}{z + 1} \right]_{z=1}$$

$$= 2\pi i (3)$$

$$= \underline{6\pi i}$$



6. Using Cauchy's integral formula evaluate $\oint_C \frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z - 1)(z - 2)} dz$ where C is $|z| = 3$.

Ans: Let $F(z) = \frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z - 1)(z - 2)}$ not analytic

$$\text{at } (z - 1)(z - 2) = 0 \Rightarrow z = 1, 2$$

The points $z = 1, 2$ lie inside the curve $|z| = 3$

$$\text{Let } \frac{1}{(z-1)(z-2)} = \frac{A}{z-1} + \frac{B}{z-2}$$

$$\Rightarrow 1 = A(z-2) + B(z-1)$$

$$\text{put } z=1, \quad 1 = A(1-2) \Rightarrow A = -1$$

$$z=2, \quad 1 = B(2-1) \Rightarrow B = 1$$

$$\Rightarrow \frac{1}{(z-1)(z-2)} = \frac{-1}{z-1} + \frac{1}{z-2}$$

$$\begin{aligned} \therefore \int_C \frac{\sin(\pi z^2) + \cos(\pi z^2)}{(z-1)(z-2)} dz &= \int_C -\left[\frac{\sin(\pi z^2) + \cos(\pi z^2)}{z-1} \right] dz \\ &\quad + \int_C \frac{\sin(\pi z^2) + \cos(\pi z^2)}{z-2} dz \\ &= -\int_C \frac{f(z)}{z-1} dz + \int_C \frac{f(z)}{z-2} dz \end{aligned}$$

where $f(z) = \sin(\pi z^2) + \cos(\pi z^2)$ analytic inside C

$$= -2\pi i f(1) + 2\pi i f(2)$$

(by Cauchy's integral formula)

$$= -2\pi i(-1) + 2\pi i(1) = \underline{\underline{4\pi i}}$$

$$\left[\begin{array}{l} \because f(1) = \sin \pi + \cos \pi = 0 - 1 = -1 \\ f(2) = \sin 4\pi + \cos 4\pi = 0 + 1 = 1 \end{array} \right]$$

7. Integrate $\frac{z^2}{z^2-1}$ counterclockwise around the circle $|z-1-i| = \pi/2$ by Cauchy's Integral formula.

Ans:

$$\text{Let } F(z) = \frac{z^2}{z^2-1} = \frac{z^2}{(z+1)(z-1)} \text{ not analytic at}$$

$$(z+1)(z-1) = 0 \Rightarrow z = 1 \text{ \& } z = -1$$

$z=1$ lies inside C

$z=-1$ lies outside C

$$\left[\begin{array}{l} \because z=1 \Rightarrow |z-1-i| = |i| = \sqrt{1} = 1 < \frac{\pi}{2} \\ z=-1 \Rightarrow |z-1-i| = |-2-i| = \sqrt{4+1} = \sqrt{5} > \frac{\pi}{2} \end{array} \right]$$

(Here $\pi = 3.14$)

$$\begin{aligned}
 \therefore \oint_C \frac{z^2}{z^2-1} dz &= \oint_C \frac{f(z)}{z-1} \quad \text{where } f(z) = \frac{z^2}{z+1} \\
 &\quad \text{analytic in } |z-1-i| = \pi/2 \\
 &= 2\pi i f(1), \text{ by Cauchy's integral formula} \\
 &= 2\pi i \left[\frac{z^2}{z+1} \right]_{z=1} \\
 &= 2\pi i \left(\frac{1}{2} \right) \\
 &= \underline{\underline{\pi i}}
 \end{aligned}$$

8. Using Cauchy's Integral formula, evaluate
 $\oint_C \frac{z^2}{z^3-z^2-z+1} dz$ where C is taken counterclockwise
 around the circle $|z+1| = 3/2$

Ans: Let $f(z) = \frac{z^2}{z^3-z^2-z+1}$

$$\begin{array}{l}
 C: |z+1| = 3/2 \\
 \text{Centre} = -1 \\
 \text{radius} = 3/2
 \end{array}$$

$$= \frac{z^2}{(z+1)(z-1)^2} \text{ not analytic at } z = \pm 1$$

The point $z = -1$ lies inside C [$\because z = -1 \Rightarrow |z+1| = |-1+1| = 0 < 3/2$]

The point $z = 1$ lies outside C [$\because z = 1 \Rightarrow |z+1| = |1+1| = 2 > 3/2$]

$$\therefore \oint_C \frac{z^2}{z^3-z^2-z+1} dz = \oint_C \frac{z^2}{(z+1)(z-1)^2} dz$$

$$= \oint \frac{f(z)}{z+1} dz \quad \text{where } f(z) = \frac{z^2}{(z-1)^2}$$

is analytic in $|z+1| = 3/2$

$$= 2\pi i f(-1), \text{ by Cauchy's integral formula}$$

$$= 2\pi i \left[\frac{z^2}{(z-1)^2} \right]_{z=-1}$$

$$= 2\pi i \left(\frac{1}{4} \right)$$

$$= \underline{\underline{\frac{\pi i}{2}}}$$

Home work :

1. Evaluate $\int \frac{3z^2 + 7z + 1}{z + 1} dz$ over the circle $|z + i| = 1$
2. Evaluate $\int \frac{4 - 3z}{z(z - 1)} dz$ over the circle $|z| = \frac{3}{2}$
3. Evaluate $\int_C \frac{\cos \pi z}{z^2 - 1} dz$ where C is the rectangle with vertices $2 \pm i$, $-2 \pm i$
4. Evaluate $\int_C \frac{z + 4}{z^2 + 2z + 5} dz$ where C is $|z + 1 - i| = 2$.

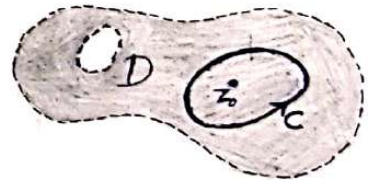
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Cauchy integral formula for derivatives of an analytic function

Theorem: If $f(z)$ is analytic in a domain D , then it has derivatives of all orders in D , which are then also analytic functions in D . The values of these derivatives at a point z_0 in D are given by the formulas

$$f'(z_0) = \frac{1}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^2} dz$$

$$f''(z_0) = \frac{2!}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^3} dz \quad \text{etc.}$$



and in general

$$f^{(n)}(z_0) = \frac{n!}{2\pi i} \oint_C \frac{f(z)}{(z-z_0)^{n+1}} dz \quad (n=1, 2, \dots); \quad (1)$$

here C is any simple closed path in D that encloses z_0 and whose full interior belongs to D ; and we integrate counterclockwise around C .

$$(1) \Rightarrow \boxed{\oint_C \frac{f(z)}{(z-z_0)^{n+1}} dz = \frac{f^{(n)}(z_0) \cdot 2\pi i}{n!}} \quad (2)$$

- 1) Evaluate $\oint_C \frac{\cos z}{(z-\pi i)^2} dz$ for any contour enclosing the point πi (counterclockwise)

Ans: Let $F(z) = \frac{\cos z}{(z-\pi i)^2}$ not analytic at $(z-\pi i)^2=0$
 $\Rightarrow z = \pi i$

$$\therefore \oint_C \frac{\cos z}{(z-\pi i)^2} dz = \oint_C \frac{f(z)}{(z-z_0)^{n+1}} \quad \left| \begin{array}{l} \text{where } f(z) = \cos z \\ \text{is analytic for all } z \\ z_0 = \pi i \text{ inside } C \\ n+1 = 2 \Rightarrow n=1 \end{array} \right.$$
$$= \frac{2\pi i f^{(n)}(z_0)}{n!}$$

$$= \frac{2\pi i f'(i)}{1!} \text{ by Cauchy's integral formula}$$

$$= 2\pi i (-i \sinh \pi)$$

$$= -2\pi i^2 \sinh \pi$$

$$= \underline{\underline{2\pi \sinh \pi}}$$

$$f(z) = \cosh z$$

$$f'(z) = \sinh z$$

$$f'(i) = \sinh(i)$$

$$= i \sinh \pi$$

$$[\sin(ix) = i \sinh x]$$

2. Evaluate $\oint_C \frac{z^4 - 3z^2 + 6}{(z+i)^3}$ where C enclosing the point $-i$ by counter clockwise integration.

Ans: Let $F(z) = \frac{z^4 - 3z^2 + 6}{(z+i)^3}$ not analytic at $(z+i)^3 = 0 \Rightarrow z = -i$

$$\begin{aligned} \therefore \oint_C \frac{z^4 - 3z^2 + 6}{(z+i)^3} &= \oint_C \frac{f(z)}{(z-z_0)^{n+1}} \\ &= \frac{2\pi i f^{(n)}(z_0)}{n!} \\ &= \frac{2\pi i f''(i)}{2!} \end{aligned}$$

$$= \frac{2\pi i (-18)}{2}$$

$$= \underline{\underline{-18\pi i}}$$

where $f(z) = z^4 - 3z^2 + 6$
analytic for all z
 $z_0 = -i$ inside C
 $n+1 = 3 \Rightarrow n = 2$

$$f(z) = z^4 - 3z^2 + 6$$

$$f'(z) = 4z^3 - 6z$$

$$f''(z) = 12z^2 - 6$$

$$f''(-i) = 12(-i)^2 - 6$$

$$= -12 - 6 = \underline{\underline{-18}}$$

3. Use Cauchy's integral formula to evaluate

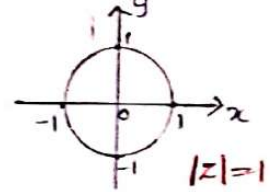
$$\int_C \frac{z+1}{z^4 + 2iz^3} dz \text{ where } C \text{ is } |z|=1$$

Ans: Let $F(z) = \frac{z+1}{z^4 + 2iz^3}$ not analytic at $z^4 + 2iz^3 = 0 \Rightarrow z^3(z + 2i) = 0$

$$\Rightarrow z^3 = 0 ; z + 2i = 0$$

$$\Rightarrow z = 0 ; z = -2i$$

Here $C : |z| = 1$



$z = 0$ lies inside & $z = -2i$ lies outside C
(0, -2)

$$\therefore \oint_C \frac{z+1}{z^4+2iz^3} dz = \oint_C \frac{z+1}{z^3(z+2i)} dz = \oint_C \frac{z+1}{z^3} dz$$

$$= \oint_C \frac{f(z)}{(z-0)^3} dz$$

$$= \frac{2\pi i}{2!} f''(0)$$

(by Cauchy's integral formula)

$$= \pi i \left(\frac{1}{2} + \frac{i}{4} \right)$$

here $f(z) = \frac{z+1}{z+2i}$
analytic in $|z|=1$
& $z=0$ inside $|z|=1$
 $n+1=3 \Rightarrow n=2$

$$\begin{aligned} \left[f(z) = \frac{z+1}{z+2i} \Rightarrow f'(z) &= \frac{(z+2i)(1) - (z+1) \cdot 1}{(z+2i)^2} = \frac{2i-1}{(z+2i)^2} \right. \\ f''(z) &= (2i-1) \frac{d}{dz} [(z+2i)^{-2}] \\ &= (2i-1) (-2)(z+2i)^{-3} = \frac{-4i+2}{(z+2i)^3} \\ \therefore f''(0) &= \frac{-4i+2}{(2i)^3} = \frac{-4i+2}{8i^3} = \frac{-4i}{8i^3} + \frac{2}{8i^3} \\ &= \frac{-4i}{-8i} + \frac{2i}{8i^4} = \frac{1}{2} + \frac{i}{4} \end{aligned}$$

A. Using Cauchy's integral formula, evaluate

$$\oint_C \frac{e^z}{(z^2+4)(z-1)^2} dz, \text{ where } C \text{ is the circle } |z-1|=2.$$

Ans: Let $F(z) = \frac{e^z}{(z^2+4)(z-1)^2}$ not analytic at

$$(z^2+4)(z-1)^2 = 0 \Rightarrow z^2+4=0 ; (z-1)^2=0$$

$$\Rightarrow z^2 = -4 ; z-1=0$$

$$\Rightarrow z = \pm 2i ; z = 1$$

Here $C: |z-1|=2$ (Circle with centre 1 & radius 2)

$z=1$ lies inside C $\left[\because z=1 \Rightarrow |z-1|=|1-1|=0 < 2 \right]$

$z=\pm 2i$ lie outside C $\left[\begin{array}{l} \because z=2i \Rightarrow |2i-1|=\sqrt{(-1)^2+2^2}=\sqrt{5}>2 \\ z=-2i \Rightarrow |-2i-1|=\sqrt{(-1)^2+(-2)^2}=\sqrt{5}>2 \end{array} \right]$

$$\begin{aligned} \therefore \oint_C \frac{e^z}{(z^2+4)(z-1)^2} dz &= \oint_C \frac{e^z / (z^2+4)}{(z-1)^2} dz \\ &= \oint_C \frac{f(z)}{(z-1)^2} dz \quad \left[\begin{array}{l} \text{where } f(z) = \frac{e^z}{z^2+4} \\ \text{analytic in } |z-1|=2 \\ z_0 = 1 \text{ inside } C \end{array} \right] \\ &= 2\pi i f'(1) \quad (\text{by Cauchy's integral formula}) \end{aligned}$$

$$= 2\pi i \left(\frac{3e}{25} \right) = \frac{6e\pi i}{25}$$

$$\left[\begin{array}{l} f(z) = \frac{e^z}{z^2+4} \quad \therefore f'(z) = \frac{(z^2+4)e^z - e^z \cdot 2z}{(z^2+4)^2} \\ \therefore f'(1) = \frac{5e - 2e}{5^2} = \frac{3e}{25} \end{array} \right]$$

5. Using Cauchy's integral formula, evaluate

$\oint_C \frac{z^2}{z^3 - z^2 - z + 1} dz$ where C is taken counter-clockwise around the circle $|z-1-i| = \pi/2$

Ans: Let $F(z) = \frac{z^2}{z^3 - z^2 - z + 1} = \frac{z^2}{(z+1)(z-1)^2}$
not analytic at $(z+1)(z-1)^2=0 \Rightarrow z=-1, 1$

Here $C: |z-(1+i)| = \pi/2$

$z=1$ lies inside C $\left[\because z=1 \Rightarrow |z-(1+i)| = |1-1-i| = |-i| = 1 < \frac{\pi}{2} \right]$

$z=-1$ lies outside C $\left[\because z=-1 \Rightarrow |-1-1-i| = |-2-i| = \sqrt{(-2)^2+(-1)^2} = \sqrt{5} > \pi/2 \right]$

$$\therefore \pi = 3.14$$

$$\therefore \oint_C \frac{z^2}{z^3 - z^2 - z + 1} dz = \oint_C \frac{z^2}{(z+1)(z-1)^2} dz = \oint_C \frac{f(z)}{(z-1)^2}$$

$$\left[\begin{array}{l} \text{where } f(z) = \frac{z^2}{z+1} \text{ analytic in } |z-1| < \frac{3}{2} \\ \text{(since } f(z) \text{ is not analytic at } z=-1 \text{ but it lies outside } C) \\ z_0 = 1, \text{ point inside } C \end{array} \right]$$

$$\frac{2\pi i}{1!} f'(1) \quad (\text{by Cauchy's integral formula})$$

$$= 2\pi i \left(\frac{3}{4} \right) = \frac{3}{2} \pi i$$

$$\left[f(z) = \frac{z^2}{z+1} \quad \therefore f'(z) = \frac{(z+1)2z - z^2(1)}{(z+1)^2} \Rightarrow f'(1) = \frac{4-1}{4} = \frac{3}{4} \right]$$

6. Evaluate $\oint_C \frac{2z^2+1}{(z+1)^2(z-2)} dz$ where $C: |z|=3$

Ans : $f(z) = \frac{2z^2+1}{(z+1)^2(z-2)}$ not analytic at $(z+1)^2(z-2)=0$
 $\Rightarrow (z+1)^2=0; z-2=0$
 $\Rightarrow z=-1; z=2$

Here $C: |z|=3$

$z=-1$ & $z=2$ both lie inside C

$$\therefore \frac{2z^2+1}{(z+1)^2(z-2)} = \frac{A}{z+1} + \frac{B}{(z+1)^2} + \frac{C}{z-2}$$

$$\Rightarrow 2z^2+1 = A(z+1)(z-2) + B(z-2) + C(z+1)^2$$

Put $z=-1, 3 = -3B \Rightarrow B = -1$

Put $z=2, 9 = 9C \Rightarrow C = 1$

Equating the co-efficient of $z^2, 2 = A+C \Rightarrow A=1$

$$\therefore \frac{2z^2+1}{(z+1)^2(z-2)} = \frac{1}{z+1} - \frac{1}{(z+1)^2} + \frac{1}{z-2}$$

$$\begin{aligned}
 \therefore \oint_C \frac{2z^2+1}{(z+1)^2(z-2)} dz &= \oint_C \frac{1}{z+1} dz + \oint_C \frac{-1}{(z+1)^2} dz + \oint_C \frac{1}{z-2} dz \\
 &= 2\pi i f(-1) - \frac{2\pi i f'(-1)}{1!} + 2\pi i f(2) \\
 &= 2\pi i (1) - 2\pi i (0) + 2\pi i (1) \\
 &= \underline{\underline{4\pi i}}
 \end{aligned}$$

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Taylor's series and Maclaurin series

Power Series: An infinite series of the form,

$$\sum_{n=0}^{\infty} a_n (z-z_0)^n = a_0 + a_1 (z-z_0) + a_2 (z-z_0)^2 + \dots$$

is called a power series about $z=z_0$

$a_0, a_1, a_2, \dots$ are real or complex constants called co-efficients of the series.

z_0 is a complex number about which the series is written is called the centre of the series.

Taylor series: Let $f(z)$ be analytic in a domain D , and let $z=z_0$ be any point in D . Then there exists precisely one Taylor series with centre z_0 that represent $f(z)$ given by

$$f(z) = f(z_0) + \frac{f'(z_0)}{1!} (z-z_0) + \frac{f''(z_0)}{2!} (z-z_0)^2 + \dots + \frac{f^{(n)}(z_0)}{n!} (z-z_0)^n + \dots$$

This representation is valid in the largest open disk with centre z_0 in which $f(z)$ is analytic.

When $z_0 = 0$, the Taylor series becomes

$$f(z) = f(0) + f'(0)z + \frac{f''(0)}{2!} z^2 + \dots + \frac{f^{(n)}(0)}{n!} z^n + \dots$$

and is called Maclaurin series.

Note: Since the Taylor's series representation of an analytic function is unique, we can obtain this representation directly expanding the function using binomial theorem for any index.

Problems :

1. Expand $\cos z$ in a Taylor's series about $z = \pi/4$

Ans: Here $f(z) = \cos z$; $f(\pi/4) = \cos \pi/4 = \frac{1}{\sqrt{2}}$

$$f'(z) = -\sin z ; f'(\pi/4) = -\sin \pi/4 = -\frac{1}{\sqrt{2}}$$

$$f''(z) = -\cos z ; f''(\pi/4) = -\cos \pi/4 = -\frac{1}{\sqrt{2}}$$

$$f'''(z) = \sin z ; f'''(\pi/4) = \sin \pi/4 = \frac{1}{\sqrt{2}}$$

$$\begin{aligned}\therefore f(z) = \cos z &= f(\pi/4) + (z - \pi/4)f'(\pi/4) + \frac{(z - \pi/4)^2}{2!}f''(\pi/4) + \dots \\ &= \frac{1}{\sqrt{2}} + (z - \pi/4)\left(-\frac{1}{\sqrt{2}}\right) + \frac{(z - \pi/4)^2}{2!}\left(-\frac{1}{\sqrt{2}}\right) + \frac{(z - \pi/4)^3}{3!}\left(\frac{1}{\sqrt{2}}\right) + \dots \\ &= \frac{1}{\sqrt{2}} \left[1 - (z - \pi/4) - \frac{(z - \pi/4)^2}{2!} + \frac{(z - \pi/4)^3}{3!} + \dots \right]\end{aligned}$$

2. Expand $f(z) = \frac{1}{z^2}$ as Taylor's series about $z = 2$

Ans:

$$f(z) = \frac{1}{z^2}, f'(z) = -\frac{2}{z^3}, f''(z) = \frac{6}{z^4}, f'''(z) = -\frac{24}{z^5} \text{ etc.}$$

$$\therefore f(2) = \frac{1}{4}, f'(2) = -\frac{2}{8} = -\frac{1}{4}, f''(2) = \frac{6}{16} = \frac{3}{8}, f'''(2) = -\frac{3}{4} \text{ etc.}$$

$$\begin{aligned}\therefore f(z) = \frac{1}{z^2} &= f(2) + (z - 2)f'(2) + \frac{(z - 2)^2}{2!}f''(2) + \frac{(z - 2)^3}{3!}f'''(2) + \dots \\ &= \frac{1}{4} + (z - 2)\left(-\frac{1}{4}\right) + \frac{(z - 2)^2}{2!}\left(\frac{3}{8}\right) + \frac{(z - 2)^3}{3!}\left(-\frac{3}{4}\right) + \dots \\ &= \frac{1}{4} \left[1 - 2\left(\frac{z - 2}{2}\right) + 3\left(\frac{z - 2}{2}\right)^2 - 4\left(\frac{z - 2}{2}\right)^3 + \dots \right]\end{aligned}$$

OR (using binomial theorem)

$$f(z) = \frac{1}{z^2} = \frac{1}{[(z - 2) + 2]^2} = \frac{1}{\left(2\left[1 + \left(\frac{z - 2}{2}\right)\right]\right)^2}$$

$$= \frac{1}{4} \left[1 + \left(\frac{z-3}{4} \right) \right]^{-1}$$

$$= \frac{1}{4} \left[1 - 2 \left(\frac{z-3}{4} \right) + 3 \left(\frac{z-3}{4} \right)^2 - 4 \left(\frac{z-3}{4} \right)^3 + \dots \right]$$

provided $|\frac{z-3}{4}| < 1$

Results For $|z| < 1$, $(1-z)^{-1} = 1 + z + z^2 + z^3 + \dots$

$$(1+z)^{-1} = 1 - z + z^2 - z^3 + \dots$$

$$(1-z)^{-2} = 1 + 2z + 3z^2 + 4z^3 + \dots$$

$$(1+z)^{-2} = 1 - 2z + 3z^2 - 4z^3 + \dots$$

3. Expand $f(z) = \frac{1}{z+1}$ as a Taylor's series about
 (i) $z=3$ (ii) $z=-2$. State the region of validity in each expansions.

Ans: (i) $f(z) = \frac{1}{z+1} = \frac{1}{(z-3)+3+1} = \frac{1}{(z-3)+4}$

$$= \frac{1}{4 \left[1 + \left(\frac{z-3}{4} \right) \right]} = \frac{1}{4} \left[1 + \left(\frac{z-3}{4} \right) \right]^{-1}$$

$$= \frac{1}{4} \left[1 - \left(\frac{z-3}{4} \right) + \left(\frac{z-3}{4} \right)^2 - \left(\frac{z-3}{4} \right)^3 + \dots \right]$$

(using binomial expansion of $(1+z)^{-1}$)

provided $|\frac{z-3}{4}| < 1$.

Thus the region of validity for the above series is $|z-3| < 4$

(ii) $f(z) = \frac{1}{z+1} = \frac{1}{(z+2)-2+1} = \frac{1}{(z+2)-1}$

$$= -\frac{1}{1-(z+2)} = -[1 - (z+2)]^{-1}$$

$$= -[1 + (z+2) + (z+2)^2 + (z+2)^3 + \dots]$$

provided $|z+2| < 1$

Thus the region of validity for the above series is $|z+2| < 1$

4. Expand $f(z) = \frac{z+1}{z-1}$ as a Taylor series about $z=-1$. State the region of validity.

Ans: $f(z) = \frac{z+1}{z-1} = 1 + \frac{2}{z-1}$ (on division)

$$= 1 + \frac{2}{(z+1)-1-1}$$

$$= 1 + \frac{2}{(z+1)-2}$$

$$= 1 + \frac{2}{(-2)\left[1 - \left(\frac{z+1}{2}\right)\right]} = 1 - \left[1 - \left(\frac{z+1}{2}\right)\right]^{-1}$$

$$= 1 - \left[1 + \left(\frac{z+1}{2}\right) + \left(\frac{z+1}{2}\right)^2 + \left(\frac{z+1}{2}\right)^3 + \dots\right]$$

$$= -\left[\left(\frac{z+1}{2}\right) + \left(\frac{z+1}{2}\right)^2 + \left(\frac{z+1}{2}\right)^3 + \dots\right]$$

provided $\left|\frac{z+1}{2}\right| < 1$

Thus the region of validity is $|z+1| < 2$

5. Find the Taylor series of $f(z) = \frac{2z^2 + 9z + 5}{z^3 + z^2 - 8z - 12}$ with centre $z_0 = 1$

Ans: $f(z) = \frac{2z^2 + 9z + 5}{z^3 + z^2 - 8z - 12} = \frac{2z^2 + 9z + 5}{(z+2)^2(z-3)}$

Consider $\frac{2z^2 + 9z + 5}{(z+2)^2(z-3)} = \frac{A}{z+2} + \frac{B}{(z+2)^2} + \frac{C}{z-3}$ — (1)

(by partial fraction)

a, $2z^2 + 9z + 5 = A(z+2)(z-3) + B(z-3) + C(z+2)^2$

put $z = 3 \Rightarrow 50 = C(25) \Rightarrow C = 2$

$z = -2 \Rightarrow -5 = B(-5) \Rightarrow B = 1$

Comparing Co-efficient of z^2 : $2 = A + C \Rightarrow A = 0$

$$\therefore f(z) = \frac{1}{(z+2)^2} + \frac{2}{z-3} \quad (\text{from (i)})$$

$$\begin{aligned} \text{Now } \frac{1}{(z+2)^2} &= \frac{1}{[(z-1)+1+2]^2} = \frac{1}{[(z-1)+3]^2} = \frac{1}{(3[1+\frac{(z-1)}{3}])^2} \\ &= \frac{1}{9} \left[1 + \left(\frac{z-1}{3} \right) \right]^{-2} \\ &= \frac{1}{9} \left[1 - 2 \left(\frac{z-1}{3} \right) + 3 \left(\frac{z-1}{3} \right)^2 - 4 \left(\frac{z-1}{3} \right)^3 + \dots \right] \\ &\quad \text{provided } |z-1| < 3 \end{aligned}$$

$$\begin{aligned} \text{Also } \frac{1}{z-3} &= \frac{1}{(z-1)+1-3} = \frac{1}{(z-1)-2} \\ &= \frac{1}{-2[1 - (\frac{z-1}{2})]} = -\frac{1}{2} \left[1 - \left(\frac{z-1}{2} \right) \right]^{-1} \\ &= -\frac{1}{2} \left[1 + \left(\frac{z-1}{2} \right) + \left(\frac{z-1}{2} \right)^2 + \left(\frac{z-1}{2} \right)^3 + \dots \right] \end{aligned}$$

Substituting these in (2)

$$\begin{aligned} f(z) &= \frac{1}{9} \left[1 - 2 \left(\frac{z-1}{3} \right) + 3 \left(\frac{z-1}{3} \right)^2 - 4 \left(\frac{z-1}{3} \right)^3 + \dots \right] \\ &\quad - \left[1 + \left(\frac{z-1}{2} \right) + \left(\frac{z-1}{2} \right)^2 + \left(\frac{z-1}{2} \right)^3 + \dots \right] \end{aligned}$$

Q) Find the Maclaurin series of the following functions

i) $\frac{1}{1-z}$ ii) e^z (iii) $\cos z$ (iv) $\sin z$

v) $\cosh z$ vi) $\sinh z$ vii) $\ln(1+z)$

$$\text{Ans: } f(z) = f(0) + \frac{f'(0)}{1!} z + \frac{f''(0)}{2!} z^2 + \frac{f'''(0)}{3!} z^3 + \dots$$

$$\text{i) } f(z) = \frac{1}{1-z} \quad \therefore f(0) = 1$$

$$f'(z) = \frac{1}{(1-z)^2} \quad ; \quad f'(0) = 1$$

$$f''(z) = \frac{2}{(1-z)^3} \quad ; \quad f''(0) = 2$$

$$f'''(z) = \frac{6}{(1-z)^4} \quad ; \quad f'''(0) = 6$$

$$\dots \dots \dots ; \dots \dots \dots$$

$$\therefore \frac{1}{1-z} = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

$$= 1 + z + z^2 + z^3 + \dots \quad \text{valid in } |z| < 1$$

ii) $f(z) = e^z$, $f'(z) = e^z$, $f''(z) = e^z$, $f'''(z) = e^z$
 $f(0) = e^0 = 1$, $f'(0) = 1$, $f''(0) = 1$, $f'''(0) = 1$ etc.

$$\therefore e^z = 1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

$$= 1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \quad \text{valid in } |z| < \infty$$

iii) $f(z) = \cos z$, $f'(z) = -\sin z$, $f''(z) = -\cos z$, $f'''(z) = \sin z$
 $f^{(4)}(z) = \cos z$, $f^{(5)}(z) = -\sin z$, $f^{(6)}(z) = -\cos z$

$$\therefore f(0) = \cos 0 = 1$$
; $f'(0) = -\sin 0 = 0$, $f''(0) = -1$, $f'''(0) = 0$
 $f^{(4)}(0) = 1$, $f^{(5)}(0) = 0$, $f^{(6)}(0) = -1$

$$\therefore \cos z = 1 + \frac{0}{1!}z + \frac{(-1)}{2!}z^2 + \frac{0}{3!}z^3 + \frac{1}{4!}z^4 + \frac{0}{5!}z^5 + \frac{(-1)}{6!}z^6 + \dots$$

$$= 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \dots \quad ; |z| < \infty$$

iv) $f(z) = \sin z = z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \dots$; $|z| < \infty$

v) $f(z) = \cosh z = \frac{e^z + e^{-z}}{2}$

$$= \frac{1}{2} \left[\left(1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \right) + \left(1 - \frac{z}{1!} + \frac{z^2}{2!} - \frac{z^3}{3!} + \dots \right) \right]$$

$$= \frac{1}{2} \left[2 + 2 \frac{z^2}{2!} + 2 \frac{z^4}{4!} + 2 \frac{z^6}{6!} + \dots \right]$$

$$= 1 + \frac{z^2}{2!} + \frac{z^4}{4!} + \frac{z^6}{6!} + \dots \quad ; |z| < \infty$$

vi) $f(z) = \sinh z = \frac{e^z - e^{-z}}{2}$

$$= \frac{1}{2} \left[\left(1 + \frac{z}{1!} + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots \right) - \left(1 - \frac{z}{1!} + \frac{z^2}{2!} - \frac{z^3}{3!} + \dots \right) \right]$$

$$= \frac{1}{2} \left[2 \frac{z}{1!} + 2 \frac{z^3}{3!} + 2 \frac{z^5}{5!} + \dots \right]$$

$$= z + \frac{z^3}{3!} + \frac{z^5}{5!} + \frac{z^7}{7!} + \dots \quad ; |z| < \infty$$

$$\begin{aligned}
 \text{vi)} \quad f(z) &= \ln(1+z) & ; \quad f(0) &= \ln(1) = 0 \\
 f'(z) &= \frac{1}{1+z} & ; \quad f'(0) &= 1 \\
 f''(z) &= \frac{-1}{(1+z)^2} & ; \quad f''(0) &= -1 \\
 f'''(z) &= \frac{2}{(1+z)^3} & ; \quad f'''(0) &= 2 \\
 f^{(4)}(z) &= \frac{-6}{(1+z)^4} & ; \quad f^{(4)}(0) &= -6 \\
 & \dots & & \dots
 \end{aligned}$$

$$\begin{aligned}
 \therefore \ln(1+z) &= 0 + \frac{1}{1!} z + \frac{(-1)}{2!} z^2 + \frac{2}{3!} z^3 + \frac{(-6)}{4!} z^4 + \dots \\
 &= \underline{z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \dots} \quad ; |z| < 1
 \end{aligned}$$

Q) Find the Maclaurin series of $f(z) = \frac{1}{1+z^2}$

Ans: $f(z) = \frac{1}{1+z^2}$

By substituting $-z^2$ for z in $\frac{1}{1-z} = 1 + z + z^2 + z^3 + \dots$

$$\underline{\underline{\frac{1}{1+z^2} = \frac{1}{1-(-z^2)} = 1 - z^2 + z^4 - z^6 + \dots \quad (|z| < 1)}}$$

Q) Find the Maclaurin series of $f(z) = \arctan z$

Ans: We have $f(z) = \arctan z = \tan^{-1}(z) \therefore f(0) = 0$

$$f'(z) = \frac{1}{1+z^2}, \quad f'(0) = 1$$

$$f''(z) = \frac{-2z}{(1+z^2)^2}; \quad f''(0) = 0$$

$$f'''(z) = \frac{(1+z^2)^{-2}(-2) + 2z \cdot 2(1+z^2) \cdot 2z}{(1+z^2)^4} \therefore f'''(0) = -2$$

$$\therefore \arctan z = \underline{z - \frac{z^3}{3} + \frac{z^5}{5} - \dots} \quad ; |z| < 1$$